

JLLS-Medical-Scoping-Review: Algorithmic Screening Pipeline for Military Grey Literature

This repository contains the programmatic screening pipeline used to isolate, standardize, and evaluate raw institutional records from the Department of War’s Joint Lessons Learned System (JLLS) for an academic scoping review.

The pipeline adheres to the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines, translating qualitative Population-Concept-Context (PCC) eligibility criteria into a reproducible, automated text-mining workflow.

Project Overview

Traditional scoping reviews encounter immense publication bias when evaluating military and operational medicine due to the exclusion of grey literature. The Joint Lessons Learned Information System (JLLS) provides primary-source data, but database extracts are dense, heterogeneous, and lack a unified narrative field. This repository provides a two-phase Python screening framework designed to:

- 1. Filter and Standardize: Isolate health-related literature from a multi-domain database extract and computationally consolidate disparate text architectures into a single narrative column.
- 2. Execute Algorithmic PCC Screening: Categorize records using automated rule-based criteria targeting Disease and Non-Battle Injuries (DNBI) within the Ukrainian Military during the Russo-Ukrainian Conflict (2020–Present).

Eligibility Criteria (PCC Framework)

The underlying rule-based algorithms are mapped directly from the review's core eligibility matrix:

Criterion	Inclusion Criteria	Exclusion Criteria
Population	Ukrainian military service members (Active-duty, deployed, reserve component)	U.S. military service members, U.S. veterans, DoD civilians or contractors, Ukrainian civilians
Concept	Disease and Non-Battle Injuries (DNBI), public health, infection, hygiene, preventable illness	Traumatic combat injuries (e.g., blast, shrapnel, gunshot wounds), animal studies, biosurveillance/bioterrorism

Criterion	Inclusion Criteria	Exclusion Criteria
Context	Russo-Ukrainian conflict occurring during or after 2020	Prior or separate conflicts (e.g., Operation Enduring Freedom, Iron Swords, GWOT)
Evidence Type	Primary empirical data, technical reports, case studies, After Action Reports (AAR)*, Observation notes*	Protocols, literature reviews (systematic/narrative), editorial or news articles, dissertations, pre-2020 data

**Note: Specific to JLLS module extractions.*

Data Pipeline Architecture

The algorithmic lifecycle processes the records using a structured funnel approach:

- **Raw Extract Collection:** Contains 1,000 initial rows across 7 system components (export_01_Jun_2026_14-13-34.xlsx).
- **Phase 1: Keyword Mining:** Executes automated text queries, filtering out 810 completely non-medical records and generating a standardized intermediate workspace file of 190 items (Medical_Scoping_Review_Screening.xlsx).
- **Phase 2: Automated PCC Screening:** Applies strict logical rules for populations, concepts (DNBI), and time constraints, discarding 175 out-of-scope files to output an audited workbook (Medical_Scoping_Review_PCC_Screened.xlsx).
- **Phase 3: Human-in-the-Loop Review:** Routes 10 clear inclusion candidates and 5 mixed-population pending entries to investigators for final manual adjudication and synthesis.

How to Run the Scripts

You can run this pipeline seamlessly inside Google Colab using the archived notebooks:

1. Prerequisites: Ensure you have python 3.x and the 'pandas' and 'openpyxl' libraries installed.
2. Data Upload: Upload the raw extraction spreadsheet using the folder pane on Colab's left sidebar.
3. Execution Phase 1: Run the foundational text-mining script to isolate terms and construct the integrated narrative column.
4. Execution Phase 2: Execute the targeted PCC logical filter to auto-populate PRISMA decision metadata fields.
5. Manual Validation: Download the resulting audited file to inspect and finalize the entries marked PENDING.